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Polyimide resin composition and molded body

Abstract

A polyimide resin composition containing a polyimide resin (A) and a polyetherimide sulfone resin (B), wherein the polyimide resin (A) contains a repeating structural unit represented by the following formula (1) and a repeating structural unit represented by the following formula (2), and a content ratio of the repeating structural unit of the formula (1) with respect to the total of the repeating structural unit of the formula (1) and the repeating structural unit of the formula (2) is 20 to 70 mol %; and a molded article containing the same. ##STR00001##

(R.sub.1 represents a divalent group having from 6 to 22 carbon atoms containing at least one alicyclic hydrocarbon structure; R.sub.2 represents a divalent chain aliphatic group having from 5 to 16 carbon atoms; and X.sub.1 and X.sub.2 each independently represent a tetravalent group having from 6 to 22 carbon atoms containing at least one aromatic ring.)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present application is a 35 U.S.C. § 371 national stage application of International patent application PCT/JP2020/044407, filed Nov. 30, 2020, which is based on and claims the benefit of priority to Japanese Application No. 2019-232103, filed Dec. 23, 2019. The entire contents of these applications are incorporated herein by reference.

TECHNICAL FIELD

(2) The present invention relates to a polyimide resin composition and a molded article.

BACKGROUND ART

(3) A polyimide resin is a useful engineering plastic that has high thermal stability, high strength

and high solvent resistance due to rigidity, resonance stabilization and firm chemical bond of the molecular chain thereof, and is being applied to a wide range of fields. A polyimide resin having crystallinity is further enhanced in the heat resistance, the strength and the chemical resistance thereof, and thus is expected for applications as alternatives of metals or the like. While a polyimide resin has high heat resistance, however, it has the problems of exhibiting no thermoplasticity and having low molding processability.

(4) Vespel (registered trademark), a highly heat-resistant resin, is known as a polyimide molding material (PTL 1). This resin is difficult to process by molding due to its very low flowability even at a high temperature, and is also disadvantageous in terms of cost because it requires molding under conditions of a high temperature and a high pressure for a prolonged period of time. In contrast to this, a resin having a melting point and flowability at a high temperature, such as a crystalline resin, may be processed by molding easily and inexpensively.

(5) Thus, a polyimide resin having thermoplasticity has been reported in recent years. Such a thermoplastic polyimide resin is excellent in molding processability in addition to the original heat resistance of the polyimide resin. The thermoplastic polyimide resin is therefore applicable to a molded article for use in an inhospitable environment to which nylon or polyester, a general purpose thermoplastic resin, is inapplicable.

(6) For example, Patent Literature 2 discloses a thermoplastic polyimide resin containing a predetermined repeating structural unit obtained by reacting a tetracarboxylic acid and/or derivative thereof containing at least one aromatic ring, a diamine containing at least one alicyclic hydrocarbon structure, and a chain aliphatic diamine.

(7) In the engineering plastics field, a technique of compounding and alloying two or more thermoplastic resins is also known for the purpose of improving physical properties, imparting functions according to the application, and the like. Patent Literature 3 discloses a thermoplastic polyimide resin containing a predetermined repeating unit, and also describes that this polyimide resin is used as a polymer alloy in combination with other resins. Patent Literature 4 discloses that a polyimide resin composition containing a polyetherimide resin and a crystalline polyimide resin containing a tetracarboxylic acid component and an aliphatic diamine component has excellent heat resistance, rigidity, and impact resistance.

CITATION LIST

Patent Literature

(8) PTL 1: JP 2005-28524 A PTL 2: WO 2013/118704 PTL 3: WO 2016/147996 PTL 4: JP 2018-70699 A

SUMMARY OF INVENTION

Technical Problem

(9) The thermoplastic polyimide resin described in Patent Literature 3 is crystalline, and has excellent heat resistance, strength, chemical resistance, and the like, but there is room for further improvement in terms of tensile properties, particularly toughness, among mechanical properties. It is considered that when toughness is improved, impact resistance, vibration control, and the like are also improved, and development can be expected to applications where importance is placed on those properties. The improvement of toughness referred to here means that elongation until break when tensile stress is applied to the molded article increases, and can be evaluated by measuring the tensile fracture strain, for example.

(10) In the examples of Patent Literature 4, the tensile modulus and the tensile elongation at break of a molded article composed of a polyimide resin composition containing a polyetherimide resin and a crystalline polyimide resin are evaluated, but a tensile break at elongation exceeding that of the crystalline polyimide resin alone was not obtained in any of the examples.

(11) An object of the present invention is to provide a polyimide resin composition and a molded article having even better tensile properties such as tensile modulus and tensile fracture strain while maintaining a high level of heat resistance, bending properties, and the like derived from the

crystalline thermoplastic polyimide resin.

Solution to Problem

(12) The present inventors have found that the aforementioned object can be attained by polyimide resin composition containing a crystalline thermoplastic polyimide resin that is combined with a particular different polyimide structural unit in a particular ratio and a polyetherimide sulfone resin.

(13) That is, the present invention relates to the following.

(14) [1] A polyimide resin composition containing a polyimide resin (A) and a polyetherimide sulfone resin (B), wherein the polyimide resin (A) contains a repeating structural unit represented by the following formula (1) and a repeating structural unit represented by the following formula (2), and a content ratio of the repeating structural unit of the formula (1) with respect to the total of the repeating structural unit of the formula (1) and the repeating structural unit of the formula (2) is 20 to 70 mol %:

(15) ##STR00002##

(16) wherein R.sub.1 represents a divalent group having from 6 to 22 carbon atoms containing at least one alicyclic hydrocarbon structure; R.sub.2 represents a divalent chain aliphatic group having from 5 to 16 carbon atoms; and X.sub.1 and X.sub.2 each independently represent a tetravalent group having from 6 to 22 carbon atoms containing at least one aromatic ring.

(17) [2] A molded article containing the polyimide resin composition according to the above [1].

Advantageous Effects of Invention

(18) The polyimide resin composition and molded article of the present invention have excellent heat resistance and bending properties and good tensile properties, and therefore are expected to be developed in applications where importance is placed on impact resistance, vibration control, and the like. For example, the polyimide resin composition and molded article of the present invention can be applied to applications such as sliding members such as gears and bearings, cutting members, structural members such as robot arms, winding coating materials such as electric wires, screws, nuts, packings, speaker diaphragms, reflectors, fifth generation (5G) mobile communication system related members, various films, and the like.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a micrograph of when a cross-section perpendicular to the flow direction (MD) of the polyimide resin composition (pellets) of Example 3 is observed by a field-emission scanning transmission electron microscope (FE-STEM).

(2) FIG. 2 is a micrograph of when a cross-section perpendicular to the MD of the pellets of Example 5 is observed by a FE-STEM.

(3) FIG. 3 is a micrograph of when a cross-section perpendicular to the MD of the pellets of Example 7 is observed by a FE-STEM.

DESCRIPTION OF EMBODIMENTS

(4) [Polyimide Resin Composition]

(5) The polyimide resin composition of the present invention contains a polyimide resin (A) and a polyetherimide sulfone resin (B), wherein the polyimide resin (A) contains a repeating structural unit represented by the following formula (1) and a repeating structural unit represented by the following formula (2), and a content ratio of the repeating structural unit of the formula (1) with respect to the total of the repeating structural unit of the formula (1) and the repeating structural unit of the formula (2) is 20 to 70 mol %:

(6) ##STR00003##

wherein R.sub.1 represents a divalent group having from 6 to 22 carbon atoms containing at least one alicyclic hydrocarbon structure; R.sub.2 represents a divalent chain aliphatic group having

from 5 to 16 carbon atoms; and X.sub.1 and X.sub.2 each independently represent a tetravalent group having from 6 to 22 carbon atoms containing at least one aromatic ring.

(7) As a result of having the above-described compositional makeup, the polyimide resin composition of the present invention has even better tensile properties, such as tensile modulus and tensile fracture strain, while maintaining a high level of heat resistance, bending properties, and the like, than when component (A) alone or component (B) alone is used.

(8) Although the reason for this is not certain, the component (A) is a crystalline thermoplastic resin, the component (B) is an amorphous thermoplastic resin, and since the mutual dispersibility of those components is high, it is considered that a resin composition and a molded article in which the component (A) or component (B) is dispersed at a nano-level are formed. Since a molded article in which the component (A) or component (B) is dispersed at the nano-level disperses stress when stress is applied, for example, it is thought that when tensile stress is applied, cracks occur in a complex manner inside the molded article and strain is alleviated at a plurality of locations, whereby toughness is improved.

(9) <Polyimide Resin (A)>

(10) The polyimide resin (A) used in the present invention contains a repeating structural unit represented by the following formula (1) and a repeating structural unit represented by the following formula (2), a content ratio of the repeating structural unit of the formula (1) with respect to the total of the repeating structural unit of the formula (1) and the repeating structural unit of the formula (2) being 20 to 70 mol %:

(11) ##STR00004##

wherein R.sub.1 represents a divalent group having from 6 to 22 carbon atoms containing at least one alicyclic hydrocarbon structure; R.sub.2 represents a divalent chain aliphatic group having from 5 to 16 carbon atoms; and X.sub.1 and X.sub.2 each independently represent a tetravalent group having from 6 to 22 carbon atoms containing at least one aromatic ring.

(12) The polyimide resin (A) used in the present invention is a crystalline thermoplastic resin, which is preferably in a powder or pellet form. The thermoplastic polyimide resin is distinguished from, for example, polyimide resins formed by closing the imide ring after shaping in a state of a polyimide precursor such as a polyamic acid and having no glass transition temperature (T_g), or polyimide resins that decompose at a temperature lower than the glass transition temperature.

(13) The repeating structural unit of formula (1) will be described in detail below.

(14) R.sub.1 represents a divalent group having from 6 to 22 carbon atoms containing at least one alicyclic hydrocarbon structure. The alicyclic hydrocarbon structure herein means a ring derived from an alicyclic hydrocarbon compound, and the alicyclic hydrocarbon compound may be either saturated or unsaturated and may be either monocyclic or polycyclic.

(15) Examples of the alicyclic hydrocarbon structure include a cycloalkane ring, such as a cyclohexane ring, a cycloalkene ring, such as cyclohexene, a bicycloalkane ring, such as a norbornane ring, and a bicycloalkene ring, such as norbornene, but the alicyclic hydrocarbon structure is not limited thereto. Among these, a cycloalkane ring is preferred, a cycloalkane ring having from 4 to 7 carbon atoms is more preferred, and a cyclohexane ring is further preferred.

(16) R.sub.1 has from 6 to 22 carbon atoms, and preferably from 8 to 17 carbon atoms.

(17) R.sub.1 contains at least one alicyclic hydrocarbon structure, and preferably from 1 to 3 alicyclic hydrocarbon structures.

(18) R.sub.1 is preferably a divalent group represented by the following formula (R1-1) or (R1-2):

(19) ##STR00005##

wherein m.sub.11 and m.sub.12 each independently represent an integer of 0-2, and preferably 0 or 1; and m.sub.13 to m.sub.15 each independently represent an integer of 0-2, and preferably 0 or 1.

(20) R.sub.1 is particularly preferably a divalent group represented by the following formula (R1-3):

(21) ##STR00006##

(22) In the divalent group represented by the formula (R1-3), the conformation of the two methylene groups with respect to the cyclohexane ring may be either cis or trans, and the ratio of cis and trans may be an arbitrary value.

(23) X.sub.1 is a tetravalent group having from 6 to 22 carbon atoms containing at least one aromatic ring. The aromatic ring may be either a monocyclic ring or a condensed ring, and examples thereof include a benzene ring, a naphthalene ring, an anthracene ring and a tetracene ring, but the aromatic ring is not limited thereto. Among these, a benzene ring and a naphthalene ring are preferred, and a benzene ring is more preferred.

(24) X.sub.1 has from 6 to 22 carbon atoms, and preferably has from 6 to 18 carbon atoms.

(25) X.sub.1 contains at least one aromatic ring, and preferably contains from 1 to 3 aromatic rings.

(26) X.sub.1 is preferably a tetravalent group represented by one of the following formulae (X-1) to (X-4):

(27) ##STR00007##

wherein R.sub.11 to R.sub.18 each independently represent an alkyl group having from 1 to 4 carbon atoms; p.sub.11 to p.sub.13 each independently represent an integer of 0-2, and preferably 0; p.sub.14, p.sub.15, p.sub.16 and p.sub.18 each independently represent an integer of 0-3, and preferably 0; p.sub.17 represents an integer of 0-4, and preferably 0; and L.sub.11 to L.sub.13 each independently represent a single bond, a carbonyl group or an alkylene group having from 1 to 4 carbon atoms.

(28) X.sub.1 is a tetravalent group having from 6 to 22 carbon atoms containing at least one aromatic ring, and therefore R.sub.12, R.sub.13, p.sub.12 and p.sub.13 in the formula (X-2) are selected in such a manner that the tetravalent group represented by the formula (X-2) has from 10 to 22 carbon atoms.

(29) Similarly, L.sub.11, R.sub.14, R.sub.15, p.sub.14 and p.sub.15 in the formula (X-3) are selected in such a manner that the tetravalent group represented by the formula (X-3) has from 12 to 22 carbon atoms, and L.sub.12, L.sub.13, R.sub.1, R.sub.17, R.sub.18, p.sub.16, p.sub.17 and p.sub.18 in the formula (X-4) are selected in such a manner that the tetravalent group represented by the formula (X-4) has from 18 to 22 carbon atoms.

(30) X.sub.1 is particularly preferably a tetravalent group represented by the following formula (X-5) or (X-6):

(31) ##STR00008##

(32) Next, the repeating structural unit of formula (2) will be described in detail below.

(33) R.sub.2 represents a divalent chain aliphatic group having from 5 to 16 carbon atoms, preferably from 6 to 14 carbon atoms, more preferably from 7 to 12 carbon atoms, and further preferably from 8 to 10 carbon atoms. The chain aliphatic group herein means a group derived from a chain aliphatic compound, and the chain aliphatic compound may be either saturated or unsaturated, and may be in the form of either linear or branched chain.

(34) R.sub.2 is preferably an alkylene group having from 5 to 16 carbon atoms, more preferably an alkylene group having from 6 to 14 carbon atoms, further preferably an alkylene group having from 7 to 12 carbon atoms, and particularly preferably an alkylene group having from 8 to 10 carbon atoms. The alkylene group may be either a linear alkylene group or a branched alkylene group, and is preferably a linear alkylene group.

(35) R.sub.2 preferably represents at least one selected from the group consisting of an octamethylene group and a decamethylene group, and particularly preferably represents an octamethylene group.

(36) X.sub.2 is defined similarly to X.sub.1 in the formula (1), and preferred embodiments thereof are also the same.

(37) The content ratio of the repeating structural unit of the formula (1) with respect to the total of the repeating structural unit of the formula (1) and the repeating structural unit of the formula (2) is 20 mol % to 70 mol %. In the case where the content ratio of the repeating structural unit of the

formula (1) is in the above range, the polyimide resin may also be sufficiently crystallized in an ordinary injection molding cycle. When the content ratio is less than 20 mol %, molding processability is deteriorated, and when the content ratio is more than 70 mol %, crystallinity is deteriorated to thereby result in deterioration in heat resistance.

(38) The content ratio of the repeating structural unit of the formula (1) with respect to the total of the repeating structural unit of the formula (1) and the repeating structural unit of the formula (2) is preferably 65 mol % or less, more preferably 60 mol % or less, and further preferably 50 mol % or less from the viewpoint of exerting high crystallinity.

(39) Among these, the content ratio of the repeating structural unit of the formula (1) with respect to the total of the repeating structural unit of the formula (1) and the repeating structural unit of the formula (2) is preferably 20 mol % or more and less than 40 mol %. When the ratio falls within this range, the crystallinity of the polyimide resin (A) increases, and it is possible to obtain a resin molded article being more excellent in heat resistance.

(40) The content ratio described above is preferably 25 mol % or more, more preferably 30 mol % or more, further preferably 32 mol % or more from the viewpoint of molding processability, and is further preferably 35 mol % or less from the viewpoint of exerting high crystallinity.

(41) The content ratio of the total of the repeating structural unit of the formula (1) and the repeating structural unit of the formula (2) with respect to the total repeating structural units constituting the polyimide resin (A) is preferably 50 to 100 mol %, more preferably 75 to 100 mol %, further preferably 80 to 100 mol %, and still further preferably 85 to 100 mol %.

(42) The polyimide resin (A) may further contain a repeating structural unit represented by the following formula (3). In this case, the content ratio of the repeating structural unit of formula (3) with respect to the total of the repeating structural unit of formula (1) and the repeating structural unit of formula (2) is preferably 25 mol % or less. The lower limit thereof is not particularly limited but needs to exceed 0 mol %.

(43) When containing the repeating structural unit of formula (3), from the viewpoint of improving heat resistance, the content ratio is preferably 5 mol % or more, and more preferably 10 mol % or more, while from the viewpoint of maintaining crystallinity, the content ratio is preferably 20 mol % or less, and more preferably 15 mol % or less.

(44) ##STR00009##

wherein R.sub.3 represents a divalent group having from 6 to 22 carbon atoms containing at least one aromatic ring; and X.sub.3 represents a tetravalent group having from 6 to 22 carbon atoms containing at least one aromatic ring.

(45) R.sub.3 is a divalent group having from 6 to 22 carbon atoms containing at least one aromatic ring. The aromatic ring may be either a monocyclic ring or a condensed ring, and examples thereof include a benzene ring, a naphthalene ring, an anthracene ring and a tetracene ring, but the aromatic ring is not limited thereto. Among these, a benzene ring and a naphthalene ring are preferred, and a benzene ring is more preferred.

(46) R.sub.3 has from 6 to 22 carbon atoms, and preferably has from 6 to 18 carbon atoms.

(47) R.sub.3 contains at least one aromatic ring, and preferably contains from 1 to 3 aromatic rings.

(48) R.sub.3 is preferably a divalent group represented by the following formula (R3-1) or (R3-2):

(49) ##STR00010##

wherein m.sub.31 and m.sub.32 each independently represent an integer of 0-2, and preferably 0 or 1; m.sub.33 and m.sub.34 each independently represent an integer of 0-2, and preferably 0 or 1; R.sub.21, R.sub.22 and R.sub.23 each independently represent an alkyl group having from 1 to 4 carbon atoms, an alkenyl group having from 2 to 4 carbon atoms or an alkynyl group having from 2 to 4 carbon atoms; p.sub.21, p.sub.22 and p.sub.23 each represent an integer of 0-4, and preferably 0; and L.sub.21 represents a single bond, a carbonyl group or an alkylene group having from 1 to 4 carbon atoms.

(50) R.sub.3 is a divalent group having from 6 to 22 carbon atoms containing at least one aromatic

ring, and therefore m.sub.31, m.sub.32, R.sub.21 and p.sub.21 in the formula (R3-1) are selected in such a manner that the divalent group represented by the formula (R3-1) has from 6 to 22 carbon atoms.

(51) Similarly, L.sub.21, m.sub.33, m.sub.34, R.sub.22, R.sub.23, p.sub.22 and p.sub.23 in the formula (R3-2) are selected in such a manner that the divalent group represented by the formula (R3-2) has from 12 to 22 carbon atoms.

(52) X.sub.3 is defined similarly to X.sub.1 in the formula (1), and preferred embodiments thereof are also the same.

(53) The end structure of the polyimide resin (A) is not particularly limited, and preferably has a chain aliphatic group having 5 to 14 carbon atoms at the end thereof.

(54) The chain aliphatic group may be either saturated or unsaturated, and may be in the form of either linear or branched chain. When the polyimide resin (A) contains the above particular group at the end thereof, it is possible to obtain a resin composition excellent in heat aging resistance.

(55) Example of the saturated chain aliphatic group having from 5 to 14 carbon atoms include an n-pentyl group, an n-hexyl group, an n-heptyl group, an n-octyl group, an n-nonyl group, an n-decyl group, an n-undecyl group, a lauryl group, an n-tridecyl group, an n-tetradecyl group, an isopentyl group, a neopentyl group, a 2-methylpentyl group, a 2-methylhexyl group, a 2-ethylpentyl group, a 3-ethylpentyl group, an isooctyl group, a 2-ethylhexyl group, a 3-ethylhexyl group, an isononyl group, a 2-ethyloctyl group, an isodecyl group, an isododecyl group, an isotridecyl group and an isotetradecyl group.

(56) Example of the unsaturated chain aliphatic group having from 5 to 14 carbon atoms include a 1-pentenyl group, a 2-pentenyl group, a 1-hexenyl group, a 2-hexenyl group, a 1-heptenyl group, a 2-heptenyl group, a 1-octenyl group, a 2-octenyl group, a nonenyl group, a decenyl group, a dodecenyl group, a tridecenyl group and a tetradecenyl group.

(57) Among these, the chain aliphatic group is preferably a saturated chain aliphatic group, and more preferably a saturated linear aliphatic group. The chain aliphatic group preferably has 6 or more carbon atoms, more preferably 7 or more carbon atoms and further preferably 8 or more carbon atoms, and preferably has 12 or less carbon atoms, more preferably 10 or less carbon atoms and further preferably 9 or less carbon atoms from the viewpoint of achievement of heat aging resistance. The chain aliphatic group may be adopted singly or in combinations of two or more.

(58) The chain aliphatic group is particularly preferably at least one selected from the group consisting of an n-octyl group, an isooctyl group, a 2-ethylhexyl group, an n-nonyl group, an isononyl group, an n-decyl group and an isodecyl group, further preferably at least one selected from the group consisting of an n-octyl group, an isooctyl group, a 2-ethylhexyl group, an n-nonyl group, and an isononyl group, and most preferably at least one selected from the group consisting of an n-octyl group, an isooctyl group, and a 2-ethylhexyl group.

(59) The polyimide resin (A) preferably contains only a chain aliphatic group having from 5 to 14 carbon atoms, besides a terminal amino group and a terminal carboxy group, at the end thereof from the viewpoint of heat aging resistance. When a group, besides the above groups, is contained at the end, the content thereof with respect to the chain aliphatic group having from 5 to 14 carbon atoms is preferably 10 mol % or less and more preferably 5 mol % or less.

(60) The content of the chain aliphatic group having from 5 to 14 carbon atoms in the polyimide resin (A) is preferably 0.01 mol % or more, more preferably 0.1 mol % or more, and further preferably 0.2 mol % or more based on the total 100 mol % of the total repeating structural units constituting the polyimide resin (A) from the viewpoint of exerting excellent heat aging resistance. Further, to ensure a sufficient molecular weight and obtain good mechanical properties, the content of the chain aliphatic group having from 5 to 14 carbon atoms in the polyimide resin (A) is, with respect to a total of 100 mol % of all the repeating structural units constituting the polyimide resin (A), preferably 10 mol % or less, more preferably 6 mol % or less, further preferably 3.5 mol % or less, still further preferably 2.0 mol % or less, and even still further preferably 1.2 mol % or less.

(61) The content of the chain aliphatic group having from 5 to 14 carbon atoms in the polyimide resin (A) can be determined by depolymerization of the polyimide resin (A).

(62) The polyimide resin (A) preferably has a melting point of 360° C. or lower and a glass transition temperature of 150° C. or higher. The melting point of the polyimide resin (A) is more preferably 280° C. or higher and further preferably 290° C. or higher from the viewpoint of heat resistance, and is preferably 345° C. or lower, more preferably 340° C. or lower, and further preferably 335° C. or lower from the viewpoint of exerting high molding processability. In addition, the glass transition temperature of the polyimide resin (A) is more preferably 160° C. or higher and more preferably 170° C. or higher from the viewpoint of heat resistance, and is preferably 250° C. or lower, more preferably 230° C. or lower, and further preferably 200° C. or lower from the viewpoint of exerting high molding processability.

(63) In addition, in the polyimide resin (A), the exothermic amount (hereinafter, also simply referred to as “exothermic amount of crystallization”) of the crystallization exothermic peak observed in melting and then cooling of the polyimide resin at a cooling rate of 20° C./min with differential scanning calorimetric measurement is preferably 5.0 mJ/mg or more, more preferably 10.0 mJ/mg or more, and further preferably 17.0 mJ/mg or more from the viewpoint of enhancement of crystallinity, heat resistance, mechanical strength, and chemical resistance. The upper limit of the exothermic amount of crystallization is not particularly limited, and is usually 45.0 mJ/mg or less.

(64) The melting point, glass transition temperature, and exothermic amount of crystallization of the polyimide resin (A) can all be measured by a differential scanning calorimeter, and specifically, can be measured by the methods described in the examples.

(65) The weight average molecular weight M_w of the polyimide resin (A) is in the range of 40,000 to 150,000, preferably 40,000 to 100,000, more preferably 42,000 to 80,000, further preferably 45,000 to 70,000, and still further preferably 45,000 to 65,000. If the weight average molecular weight M_w of the polyimide resin (A) is 40,000 or more, a microphase-separated structure is easily formed in the resin molded article, the HDT in a low load environment is improved, and the mechanical strength is also good. Further, if M_w is 150,000 or less, the molding processability is good.

(66) The weight average molecular weight M_w of the polyimide resin (A) can be measured by a gel permeation chromatography (GPC) method using polymethyl methacrylate (PMMA) as a standard sample.

(67) The logarithmic viscosity of the polyimide resin (A) at 30° C. in a 5 mass % concentrated sulfuric acid solution is preferably 0.8 to 2.0 dL/g and more preferably 0.9 to 1.8 dL/g. If the logarithmic viscosity is 0.8 dL/g or more, it is easy to form a microphase-separated structure in the obtained molded article, and sufficient mechanical strength can be obtained. If the logarithmic viscosity is 2.0 dL/g or less, molding processability and handleability are good. The logarithmic viscosity μ is obtained according to the following expression by measuring the elapsed times for flowing concentrated sulfuric acid and the polyimide resin solution at 30° C. with a Cannon-Fenske viscometer. $\mu = \ln(t_s/t_{\text{sub.0}})/C$ $t_{\text{sub.0}}$: elapsed time for flowing concentrated sulfuric acid t_s : elapsed time for flowing polyimide resin solution C: 0.5 (g/dL)

(Method for Producing Polyimide Resin (A))

(68) The polyimide resin (A) may be produced by reacting a tetracarboxylic acid component and a diamine component. The tetracarboxylic acid component contains a tetracarboxylic acid containing at least one aromatic ring and/or a derivative thereof, and the diamine component contains a diamine containing at least one alicyclic hydrocarbon structure and a chain aliphatic diamine.

(69) The tetracarboxylic acid containing at least one aromatic ring is preferably a compound having four carboxy groups that are bonded directly to the aromatic ring, and may contain an alkyl group in the structure thereof. The tetracarboxylic acid preferably has from 6 to 26 carbon atoms.

Preferred examples of the tetracarboxylic acid include pyromellitic acid, 2,3,5,6-

toluenetetracarboxylic acid, 3,3',4,4'-benzophenonetetracarboxylic acid, 3,3',4,4'-biphenyltetracarboxylic acid and 1,4,5,8-naphthalenetetracarboxylic acid.

(70) Among these, pyromellitic acid is more preferred.

(71) Examples of the derivative of the tetracarboxylic acid containing at least one aromatic ring include an anhydride and an alkyl ester compound of a tetracarboxylic acid containing at least one aromatic ring. The derivative of the tetracarboxylic acid preferably has from 6 to 38 carbon atoms. Examples of the anhydride of the tetracarboxylic acid include pyromellitic monoanhydride, pyromellitic dianhydride, 2,3,5,6-toluenetetracarboxylic dianhydride, 3,3',4,4'-diphenylsulfonetetracarboxylic dianhydride, 3,3',4,4'-benzophenonetetracarboxylic dianhydride, 3,3',4,4'-biphenyltetracarboxylic dianhydride and 1,4,5,8-naphthalenetetracarboxylic dianhydride. Examples of the alkyl ester compound of the tetracarboxylic acid include dimethyl pyromellitate, diethyl pyromellitate, dipropyl pyromellitate, diisopropyl pyromellitate, dimethyl 2,3,5,6-toluenetetracarboxylate, dimethyl 3,3',4,4'-diphenylsulfonetetracarboxylate, dimethyl 3,3',4,4'-benzophenonetetracarboxylate, dimethyl 3,3',4,4'-biphenyltetracarboxylate and dimethyl 1,4,5,8-naphthalenetetracarboxylate. The alkyl group in the alkyl ester compound of the tetracarboxylic acid preferably has from 1 to 3 carbon atoms.

(72) The tetracarboxylic acid containing at least one aromatic ring and/or the derivative thereof may be used as a sole compound selected from the aforementioned compounds or may be used as a combination of two or more compounds.

(73) The diamine containing at least one alicyclic hydrocarbon structure preferably has from 6 to 22 carbon atoms, and preferred examples thereof include 1,2-bis(aminomethyl)cyclohexane, 1,3-bis(aminomethyl)cyclohexane, 1,4-bis(aminomethyl)cyclohexane, 1,2-cyclohexanediamine, 1,3-cyclohexanediamine, 1,4-cyclohexanediamine, 4,4'-diaminodicyclohexylmethane, 4,4'-methylenebis(2-methylcyclohexylamine), carvone diamine, limonene diamine, isophorone diamine, norbornane diamine, bis(aminomethyl)tricyclo[5.2.1.0.sup.2,6]decane, 3,3'-dimethyl-4,4'-diaminodicyclohexylmethane and 4,4'-diaminodicyclohexylpropane. These compounds may be used solely or may be used as a combination of two or more compounds selected therefrom. Among these, 1,3-bis(aminomethyl)cyclohexane is preferably used. A diamine containing an alicyclic hydrocarbon structure generally has conformational isomers, and the ratio of the cis isomer and the trans isomer is not particularly limited.

(74) The chain aliphatic diamine may be in the form of either linear or branched chain, and has preferably from 5 to 16 carbon atoms, more preferably from 6 to 14 carbon atoms and further preferably from 7 to 12 carbon atoms. The linear moiety having from 5 to 16 carbon atoms may contain an ether bond in the course thereof. Preferred examples of the chain aliphatic diamine include 1,5-pentamethylenediamine, 2-methylpentane-1,5-diamine, 3-methylpentane-1,5-diamine, 1,6-hexamethylenediamine, 1,7-heptamethylenediamine, 1,8-octamethylenediamine, 1,9-nonamethylenediamine, 1,10-decamethylenediamine, 1,11-undecamethylenediamine, 1,12-dodecamethylenediamine, 1,13-tridecamethylenediamine, 1,14-tetradecamethylenediamine, 1,16-hexadecamethylenediamine, and 2,2'-(ethylenedioxy)bis(ethyleneamine).

(75) The chain aliphatic diamine may be used as a sole compound or as a mixture of plural kinds thereof. Among these, a chain aliphatic diamine having from 8 to 10 carbon atoms can be preferably used, and at least one selected from the group consisting of 1,8-octamethylenediamine and 1,10-decamethylenediamine can be particularly preferably used.

(76) In the production of the polyimide resin (A), the molar ratio of the charged amount of the diamine containing at least one alicyclic hydrocarbon structure with respect to the total amount of the diamine containing at least one alicyclic hydrocarbon structure and the chain aliphatic diamine is preferably 20 to 70 mol %. The molar ratio is preferably 25 mol % or more, more preferably 30 mol % or more, further preferably 32 mol % or more, and is preferably 60 mol % or less, more preferably 50 mol % or less, further preferably less than 40 mol, and further preferably 35 mol % or less from the viewpoint of exerting high crystallinity.

- (77) The diamine component may contain a diamine containing at least one aromatic ring. The diamine containing at least one aromatic ring preferably has from 6 to 22 carbon atoms, and examples thereof include o-xylylenediamine, m-xylylenediamine, p-xylylenediamine, 1,2-diethynylbenzenediamine, 1,3-diethynylbenzenediamine, 1,4-diethynylbenzenediamine, 1,2-diaminobenzene, 1,3-diaminobenzene, 1,4-diaminobenzene, 4,4'-diaminodiphenylmethane, α,α' -bis(4-aminophenyl)-1,4-diisopropylbenzene, α,α' -bis(3-aminophenyl)-1,4-diisopropylbenzene, 2,2-bis(4-(4-aminophenoxy)phenyl)propane, 2,6-diaminonaphthalene and 1,5-diaminonaphthalene.
- (78) The molar ratio of the charged amount of the diamine containing at least one aromatic ring with respect to the total amount of the diamine containing at least one alicyclic hydrocarbon structure and the chain aliphatic diamine is preferably 25 mol % or less. The lower limit thereof is not particularly limited but needs to exceed 0 mol %.
- (79) The molar ratio is preferably 5 mol % or more, and more preferably 10 mol % or more, from the viewpoint of enhancement of the heat resistance, and is preferably 20 mol % or less, and more preferably 15 mol % or less, from the viewpoint of maintenance of the crystallinity.
- (80) In addition, the molar ratio is preferably 12 mol % or less, more preferably 10 mol % or less, further preferably 5 mol % or less and still more preferably 0 mol % from the viewpoint of a decrease in coloration of the polyimide resin.
- (81) In the production of the polyimide resin (A), the charged amount ratio of the tetracarboxylic acid component and the diamine component is preferably from 0.9 to 1.1 mol of the diamine component per 1 mol of the tetracarboxylic acid component.
- (82) In the production of the polyimide resin (A), an end capping agent may be mixed in addition to the tetracarboxylic acid component and the diamine component. The end capping agent is preferably at least one selected from the group consisting of a monoamine compound and a dicarboxylic acid compound. It is sufficient for the amount of the end capping agent used to be an amount in which a desired amount of the end group can be introduced into the polyimide resin (A). This used amount is, based on one mole of the tetracarboxylic acid and/or derivative thereof, preferably from 0.0001 to 0.1 moles, more preferably from 0.001 to 0.06 moles, further preferably from 0.002 to 0.035 moles, still further preferably from 0.002 to 0.020 moles, and even still further preferably from 0.002 to 0.012 moles.
- (83) Among them, monoamine end capping agents are preferable as the end capping agent, and from the viewpoint of introducing the above-described chain aliphatic group having 5 to 14 carbon atoms at an end of the polyimide resin (A) to improve heat aging resistance, a monoamine that has a chain aliphatic group having 5 to 14 carbon atoms is more preferable, and a monoamine that has a saturated linear aliphatic group having 5 to 14 carbon atoms is further preferable.
- (84) The end capping agent is particularly preferably at least one selected from the group consisting of n-octylamine, isooctylamine, 2-ethylhexylamine, n-nonylamine, isononylamine, n-decylamine, and isodecylamine, further preferably at least one selected from the group consisting of n-octylamine, isooctylamine, 2-ethylhexylamine, n-nonylamine, and isononylamine, and most preferably at least one selected from the group consisting of n-octylamine, isooctylamine, and 2-ethylhexylamine.
- (85) As a polymerization method for producing the polyimide resin (A), a known polymerization method may be applied, and the method described in WO 2016/147996 can be used.
- (86) <Polyetherimide Sulfone Resin (B)>
- (87) The polyimide resin composition of the present invention contains the polyimide resin (A) and the polyetherimide sulfone resin (B). By compounding the polyimide resin (A) and the polyetherimide sulfone resin (B), a polyimide resin composition and a molded article having better tensile properties while maintaining a high level of heat resistance, bending properties, and the like can be produced.
- (88) The polyetherimide sulfone resin used as the component (B) is an amorphous thermoplastic resin containing a repeating structural unit having an imide structure, an ether bond, and —

SO.sub.2—.

(89) From the viewpoint of obtaining good heat resistance and tensile properties by compounding with the polyimide resin (A), the glass transition temperature of the polyetherimide sulfone resin (B) is preferably 230° C. or higher, more preferably 235° C. or higher, and further preferably 240° C. or higher. Further, from the viewpoint of molding processability, the glass transition temperature is preferably 280° C. or lower, and more preferably 260° C. or lower.

(90) The glass transition temperature can be measured by the same method as described above.

(91) Examples of the polyetherimide sulfone resin (B) include polyetherimide sulfone resins containing a repeating structural unit represented by the following formula (4).

(92) ##STR00011##

wherein R.sub.4 represents a divalent group having from 6 to 22 carbon atoms containing —SO.sub.2—, and X.sub.4 represents a tetravalent group having from 12 to 40 carbon atoms containing at least one aromatic ring and an ether bond.

(93) R.sub.4 represents a divalent group having from 6 to 22 carbon atoms containing —SO.sub.2—, and from the viewpoint of obtaining good heat resistance and tensile properties, R.sub.4 preferably contains at least one aromatic ring or alicyclic hydrocarbon structure. The definition of the aromatic ring and alicyclic hydrocarbon structure is the same as described above.

(94) More preferably, R.sub.4 represents a divalent group having from 6 to 22 carbon atoms containing —SO.sub.2— and at least one aromatic ring, and further preferably a divalent group having from 6 to 22 carbon atoms containing —SO.sub.2 and 1 to 3 aromatic rings.

(95) R.sub.4 has from 6 to 22 carbon atoms, preferably from 6 to 18 carbon atoms, and more preferably from 12 to 18 carbon atoms.

(96) R.sub.4 preferably represents a divalent group represented by the following formula (R4-1).

(97) ##STR00012##

wherein R.sub.41 and R.sub.42 are each independently an alkyl group having from 1 to 4 carbon atoms, an alkenyl group having from 2 to 4 carbon atoms, or an alkynyl group having from 2 to 4 carbon atoms, m.sub.41 and m.sub.42 are each independently an integer of from 0 to 2, and p.sub.41 and p.sub.42 are each independently an integer of from 0 to 4.

(98) R.sub.41 and R.sub.42 are preferably alkyl groups having from 1 to 4 carbon atoms, and more preferably are methyl groups.

(99) m.sub.41 and m.sub.42 are preferably 0 or 1, and more preferably 0.

(100) p.sub.41 and p.sub.42 are preferably from 0 to 2, and more preferably 0.

(101) X.sub.4 is a tetravalent group that contains at least one aromatic ring and an ether bond and that has from 12 to 40 carbon atoms.

(102) X.sub.4 preferably has from 18 to 40 carbon atoms, and more preferably from 24 to 30 carbon atoms.

(103) X.sub.4 contains at least one aromatic ring, and preferably 2 to 4 aromatic rings. Further, X.sub.4 contains at least one ether bond, and preferably 2 to 4 ether bonds.

(104) X.sub.4 is preferably a tetravalent group represented by the following formula (X-7).

(105) ##STR00013##

wherein L.sub.41 is a divalent group having from 4 to 28 carbon atoms.

(106) The divalent group in L.sub.41 may be a chain aliphatic group or may be a group containing an aromatic ring or an alicyclic hydrocarbon structure. From the viewpoint of obtaining good heat resistance and tensile properties, L.sub.41 preferably contains at least one aromatic ring, and more preferably contains 1 to 2 aromatic rings.

(107) L.sub.41 has from 4 to 28 carbon atoms, preferably from 6 to 24 carbon atoms, and more preferably from 12 to 18 carbon atoms.

(108) More preferably, L.sub.41 represents a divalent group represented by any of the following formulas (X-7a) to (X-7e), and further preferably represents a divalent group represented by the following formula (X-7a).

(109) ##STR00014##

wherein R.sub.43 represents an alkyl group having from 1 to 4 carbon atoms, and p.sub.43 represents an integer of from 0 to 4.

(110) R.sub.43 preferably represents a methyl group, and p.sub.43 preferably represents an integer of from 0 to 2, and more preferably 0.

(111) More preferably, from the viewpoint of obtaining good heat resistance and tensile properties, the polyetherimide sulfone resin (B) contains the repeating structural unit represented by the following formula (4-1).

(112) ##STR00015##

(113) The polyetherimide sulfone resin containing the repeating structural unit represented by formula (4-1) is obtained, for example, by reacting 2,2-bis[4-(2,3-dicarboxyphenoxy)phenyl]propane dianhydride, which is a tetracarboxylic acid component, and diaminodiphenyl sulfone, which is a diamine component.

(114) One polyetherimide sulfone resin (B) may be used alone, or two or more may be used in combination.

(115) Commercially available products can also be used as the polyetherimide sulfone resin (B). Examples of commercially available polyetherimide sulfone resins include the EXTEM VH series produced by Sabic Innovative Plastics.

(116) The mass ratio of the polyimide resin (A) to the polyetherimide sulfone resin (B), [(A)/(B)], in the polyimide resin composition of the present invention is preferably in the range of 1/99 to 99/1, and more preferably 5/95 to 95/5.

(117) In this range, the range in which the toughness of the obtained molded article is particularly excellent is the range in which the mass ratio [(A)/(B)] is 25/75 to 95/5, preferably 30/70 to 95/5, more preferably 45/55 to 95/5, further preferably 50/50 to 90/10, still further preferably 55/45 to 85/15, and even still further preferably 60/40 to 80/20.

(118) On the other hand, when importance is placed on molding processability, the mass ratio [(A)/(B)] is preferably in the range of 65/35 to 99/1, and from the viewpoint of molding processability and the toughness of the obtained molded article, more preferably in the range of 65/35 to 95/5, and further preferably 70/30 to 95/5.

(119) Further, from the viewpoint of obtaining the effects of the present invention, the total content of the polyimide resin (A) and the polyetherimide sulfone resin (B) in the polyimide resin composition is preferably 30% by mass or more, more preferably 50% by mass or more, further preferably 70% by mass or more, still further preferably 80% by mass or more, and even still further preferably 90% by mass or more. The upper limit is 100% by mass.

(120) <Reinforcement Fiber>

(121) The polyimide resin composition of the present invention may further contain a reinforcement fiber for the purpose of improving the tensile modulus among the tensile properties. Examples of the reinforcement fiber include glass fiber, carbon fiber, aramid fiber, boron fiber, metal fiber, and the like. One type of reinforcement fiber may be used alone, or two or more may be used in combination.

(122) There is no particular limitation on the form of the reinforcement fiber, and depending on the form of the obtained polyimide resin composition and molded article, either a short fiber or a continuous fiber may be used, or both may be used in combination.

(123) The form of the polyimide resin composition will be described later, but for example when the polyimide resin composition is a pellet, the reinforcement fiber is preferably a short fiber from the viewpoint of extrusion moldability and the like. The polyimide resin composition may be in the form of a prepreg in which a mixture containing the component (A) and the component (B) is impregnated in the reinforcement fiber. In this case the reinforcement fiber is preferably a continuous fiber.

(124) If the reinforcement fiber is a short fiber, from the viewpoints of strength and handleability,

the average fiber length is preferably 0.2 mm or more, more preferably 0.5 mm or more, and further preferably 2 mm or more, and from the viewpoint of extrusion moldability and the like, the average fiber length is preferably 25 mm or less, more preferably 15 mm or less, and further preferably less than 10 mm.

(125) If the reinforcement fiber is continuous fibers, examples thereof include various forms of the reinforcement fiber in which monofilaments or multifilaments are simply arranged in one direction or alternately, fabrics such as woven fabrics, non-woven fabrics, mats, and the like. Among these, the reinforcement fiber in the form of monofilaments, fabric, non-woven fabric, or a mat is preferable, and in the form of fabric is more preferable.

(126) If the reinforcement fiber is continuous fibers, the fineness of the fibers is preferably 20 to 4,500 tex, and more preferably 50 to 4,000 tex. When the fineness is in this range, impregnation of the resin component is easy, and the elastic modulus and strength of the obtained molded article are excellent. The fineness can be calculated by determining the weight of a continuous fiber of any length and converting into a weight per 1,000 m.

(127) The average fiber diameter of the reinforcement fiber is preferably 1 to 100 μm , more preferably 3 to 50 μm , and further preferably 4 to 20 μm . When the average fiber diameter is in this range, processing is easy, and the elastic modulus and strength of the obtained molded article are excellent.

(128) The average fiber length of the reinforcement fiber (short fibers) and the average fiber diameter of the reinforcement fiber are obtained by randomly selecting and observing 50 or more fibers with a scanning electron microscope (SEM) or the like, measuring, and calculating the number average.

(129) Among reinforcement fibers, carbon fiber is preferable from the viewpoint of improving the tensile modulus and lightness. Examples of the carbon fiber include polyacrylonitrile-based carbon fiber and pitch-based carbon fiber.

(130) The number of filaments of the carbon fiber is usually in the range of 500 to 100,000, preferably 5,000 to 80,000, more preferably 10,000 to 70,000.

(131) In order to improve the wettability and interface adherence with the polyimide resin (A), it is preferable that the carbon fiber is surface treated with a surface treatment agent. The surface treatment agent is a concept that includes a bundling agent and a sizing agent.

(132) Examples of the surface treatment agent include epoxy-based materials, urethane-based materials, acrylic materials, polyamide-based materials, polyimide-based materials, polyester-based materials, vinyl ester-based materials, polyolefin-based materials, polyether-based materials, and the like. One or more of these can be used in combination. From the viewpoint of obtaining higher mechanical properties, an epoxy-based material or a polyimide-based material is preferable as the surface treatment agent.

(133) Commercially available products can also be used as the carbon fiber. Examples of commercially available carbon fiber (short fibers) include the chopped fibers “CFUW”, “CFEPP”, “CFEPU”, “CFA4”, “FX1”, “EX1”, “BF-WS”, “CF-N”, and “VX-1” series produced by Nihon Polymer Sangyo Co., Ltd., the “Pyrofil Chopped Fiber” series produced by Mitsubishi Chemical Corporation, and the “Tenax-J” series (HT C702, IM C702, etc.) and the “Tenax-A” series (IM P303, HT P722, etc.) produced by Teijin Limited.

(134) If the polyimide resin composition of the present invention contains a reinforcement fiber, the content of the reinforcement fiber in the polyimide resin composition is preferably 1 to 70% by mass, more preferably 5 to 50% by mass, and further preferably 5 to 35% by mass. If the content of the reinforcement fiber in the polyimide resin composition is 1% by mass or more, an effect of improving tensile modulus can be obtained, and if the content is 70% by mass or less, molding processability, and the thermal deformation temperature (HDT) of the obtained molded article and the like are easily maintained.

(135) In general, it is known that for amorphous resins, the effect of an improvement in physical

properties by containing a reinforcement fiber is low. However, for the polyimide resin composition of the present invention, even if it contains the component (B), which is an amorphous resin, the effect of an improvement on tensile modulus and thermal deformation temperature due to the reinforcement fiber is high.

(136) <Additives>

(137) The polyimide resin composition of the present invention may contain additives such as a filler other than the reinforcement fiber, a delusterant, a nucleating agent, a plasticizer, an antistatic agent, an anti-coloring agent, an anti-gelling agent, a flame retardant, a colorant, a slidability improver, an antioxidant, a UV absorbent, a conducting agent, and a resin-modifying agent, as necessary.

(138) The content of the additive is not particularly limited, but from the viewpoint that the effect of the additive is exerted while maintaining the physical properties derived from the polyimide resin (A) and the polyetherimide sulfone resin (B), the content in the polyimide resin composition is usually 50% by mass or less, preferably from 0.0001 to 30% by mass, more preferably from 0.0001 to 15% by mass, and further preferably from 0.001 to 10% by mass.

(139) Although the polyimide resin composition of the present invention can take any form, it is preferably a pellet.

(140) Since the polyimide resin (A) and the polyetherimide sulfone resin (B) have thermoplasticity, for example, the polyimide resin (A), the polyetherimide sulfone resin (B), and various optional components as necessary can be melt-kneaded in an extruder to extrude a strand, and the strand can then be cut into pellets. A molded article having a desired shape can be easily produced by introducing the obtained pellets into a molding machine and thermoforming by the method described later.

(141) The glass transition temperature of the polyimide resin composition of the present invention is, from the viewpoint of heat resistance, preferably 160° C. or higher, and more preferably 170° C. or higher, and from the viewpoint of exerting high molding processability, the glass transition temperature is preferably 250° C. or lower, more preferably 240° C. or lower, further preferably 230° C. or lower, and still further preferably 200° C. or lower. The glass transition temperature can be measured by the same method as described above.

(142) <Tensile Properties>

(143) According to the polyimide resin composition of the present invention, a molded article can be provided that has better tensile properties than in the case of the polyimide resin (A) alone or the polyetherimide sulfone resin (B).

(144) For example, for tensile fracture strain, using a 1A-type test piece specified in JIS K7161-2: 2014 obtained by molding the polyimide resin composition, the tensile fracture strain measured by carrying out a tensile test at a temperature of 23° C., a distance between grips of 50 mm, and a test speed of 5 mm/min in accordance with JIS K7161-1: 2014 and K7161-2: 2014, can be preferably 25% or more, more preferably 40% or more, further preferably 50% or more, and still further preferably 70% or more. Specifically, the tensile fracture strain can be measured by the method described in the examples.

(145) <Bending Properties>

(146) As described above, the polyimide resin composition of the present invention can have improved tensile properties while maintaining a high level of bending properties. For the bending properties, using the molded article of 80 mm×10 mm×thickness 4 mm specified in ISO 316 obtained by molding the polyimide resin composition, a bending strength and a bending modulus, measured by carrying out a bending test at a temperature of 23° C. and a test speed of 2 mm/min in accordance with ISO 178: 2010, can be 100 MPa or more and 2.2 GPa or more, respectively. Specifically, the bending strength and the flexural modulus can be measured by the methods described in the examples.

(147) <Other Properties>

(148) According to the polyimide resin composition of the present invention, a molded article having a pearly luster with a higher degree of whiteness than in the case of the polyetherimide sulfone resin (B) alone can be produced. Therefore, the polyimide resin composition of the present invention and a molded article thereof are expected to also be applied to reflectors and the like. Further, since the polyimide resin composition of the present invention has a performance derived from the polyimide resin (A), which is a crystalline thermoplastic resin, chemical resistance is also good.

(149) [Molded Article]

(150) The present invention provides a molded article including the polyimide resin composition.

(151) Since the polyimide resin composition of the present invention has thermoplasticity, the molded article of the present invention can be easily produced by heat-molding. Examples of the heat molding method include injection molding, extrusion molding, blow molding, heat press molding, vacuum molding, pneumatic molding, laser molding, welding, and heat adhesion, and the polyimide resin composition of the present invention may be molded by any molding method that includes a heat melting step.

(152) The molding temperature depends on the thermal properties (melting point and glass transition temperature) of the polyimide resin composition, but for example, in injection-molding, molding is possible at a molding temperature of lower than 400° C. and a mold temperature of 220° C. or lower.

(153) The method for producing a molded article preferably includes the step of heat-molding the polyimide resin composition at a temperature of lower than 400° C. Examples of the specific procedure include the following methods.

(154) First, the polyimide resin (A) is dry blended with the polyetherimide sulfone resin (B) and various optional components as necessary, then introduced into an extruder, melted at a temperature of preferably less than 400° C., and melt-kneaded in the extruder and extruded to produce pellets. Alternatively, the polyimide resin (A) may be introduced into the extruder, melted at a temperature of preferably less than 400° C., the polyetherimide sulfone resin (B) and various optional components introduced thereto and melt-kneaded with the polyimide resin (A) in the extruder, and extruded to produce the pellets.

(155) The pellets may be dried, then introduced in various kinds of molding machines, and heat-molded preferably at a temperature of lower than 400° C., thereby producing a molded article having a desired shape.

(156) The molded article of the present invention has excellent heat resistance and bending properties and good tensile properties, and therefore is expected to be developed in applications where importance is placed on impact resistance, vibration control, and the like. For example, the molded article of the present invention can be applied to applications such as sliding members such as gears and bearings, cutting members, structural members such as robot arms, winding coating materials such as electric wires, screws, nuts, packings, speaker diaphragms, reflectors, fifth generation (5G) mobile communication system related members, various films, and the like.

EXAMPLES

(157) The present invention will be described in more detail with reference to examples below, but the present invention is not limited thereto. Further, various measurements and evaluations in each Production Example and Example were carried out in the following manner.

(158) <Infrared Spectroscopy (IR Measurement)>

(159) The IR measurement of the polyimide resin was performed with “JIR-WINSPEC 50”, produced by JEOL, Ltd.

(160) <Logarithmic Viscosity μ >

(161) The polyimide resin was dried at from 190 to 200° C. for 2 hours, and then 0.100 g of the polyimide resin was dissolved in 20 mL of concentrated sulfuric acid (96%, produced by Kanto Chemical Co., Inc.) to form a polyimide resin solution, and the measurement was made at 30° C.

with a Cannon-Fenske viscometer using the polyimide resin solution as a measurement sample. The logarithmic viscosity η was obtained according to the following expression. $\eta = \ln(t_s/t_{sub.0})/C$
t.sub.0: elapsed time for flowing concentrated sulfuric acid t_s: elapsed time for flowing polyimide resin solution C: 0.5 g/dL

<Melting Point, Glass Transition Temperature, Crystallization Temperature, and Exothermic Amount of Crystallization>

(162) The melting point T_m, glass transition temperature T_g, crystallization temperature T_c, and the exothermic amount of crystallization ΔH_m of the polyimide resin and the polyimide resin composition produced in each of the examples were measured with a differential scanning calorimeter (“DSC-6220”, produced by SII Nanotechnology, Inc.).

(163) In a nitrogen atmosphere, a thermal history of the following conditions was imposed on the polyimide resin or polyimide resin composition. The condition of the thermal history included the first heating (heating rate: 10° C./min), then cooling (cooling rate: 20° C./min), and then second heating (heating rate: 10° C./min).

(164) The melting point T_m was determined by reading the peak top value of the endothermic peak observed in the second heating. The glass transition temperature (T_g) was determined by reading the value observed in the second heating. The crystallization temperature (T_c) was determined by reading the peak top value of the exothermic peak observed in cooling. For T_m, T_g, and T_c, in cases where a plurality of the peaks were observed, the peak top value of the peaks was read.

(165) The exothermic amount of crystallization ΔH_m (mJ/mg) was calculated from the area of the exothermic peak observed in cooling.

(166) <Crystallization Half-Time>

(167) The crystallization half-time of the polyimide resin was measured with a differential scanning calorimeter (“DSC-6220”, produced by SII Nanotechnology, Inc.).

(168) The polyimide resin was held at 420° C. for 10 minutes in a nitrogen atmosphere so as to completely melt, then quenched at a cooling rate of 70° C./min, and the time required from the appearance of the observed crystallization peak to the peak top thereof was calculated. In Table 1, cases where the crystallization half-time was 20 seconds or less are indicated as “<20”.

(169) <Weight Average Molecular Weight>

(170) The weight average molecular weight (M_w) of the polyimide resin was measured with a gel permeation chromatography (GPC) measurement apparatus “Shodex GPC-101” produced by Showa Denko K.K. under the following conditions: Column: Shodex HFIP-806M Mobile phase solvent: HFIP containing 2 mM sodium trifluoroacetate Column temperature: 40° C. Flow rate of mobile phase: 1.0 mL/min Specimen concentration: about 0.1 mass % Detector: IR detector Amount of injection: 100 μ m Calibration curve: standard PMMA

<Bending Strength and Flexural Modulus>

(171) Using the polyimide resin, polyethersulfone resin, or polyimide resin composition produced in each of the examples, molded articles of 80 mm×10 mm×thickness 4 mm specified in ISO 316 were produced by the method described later, and used for measurement. The bending test was performed with Bend Graph (produced by Toyo Seiki Seisaku-Sho, Ltd.) according to ISO 178: 2010 at a temperature of 23° C. and a testing speed of 2 mm/min to measure the bending strength and the flexural modulus.

(172) <Tensile Strength, Tensile Modulus, and Tensile Fracture Strain>

(173) Using the polyimide resin, polyethersulfone resin, or polyimide resin composition produced in each of the examples, 1A-type test pieces specified in JIS K7161-2: 2014 were produced and used for measurement by the methods described later. Using a tensile tester (Strograph VG-1E, produced by Toyo Seiki Co., Ltd.), a tensile test was performed at a temperature of 23° C., a distance between grips of 50 mm, and a test speed of 5 mm/min in accordance with JIS K7161-1: 2014 and K7161-2: 2014, and the tensile strength, tensile modulus, and tensile fracture strain were measured.

(174) <Color Hue>

(175) Pellets of each of the polyimide resin, polyethersulfone resin, or polyimide resin composition produced in each of the examples were used for measurement.

(176) The Lab value and YI value were measured by a reflection method using a color difference meter ("ZE2000", produced by Nippon Denshoku Industries Co., Ltd.). Further, the degree of whiteness was calculated based on the Lab value and the YI value.

(177) The Lab value was measured in accordance with JIS Z8781-4:2013, the YI value was measured in accordance with JIS K7373:2006, and the degree of whiteness was calculated in accordance with JIS Z8715:1999.

(178) <Storage Modulus, Loss Modulus, and Loss Tangent (Tan δ)>

(179) A 1A-type test piece (1 mm thickness) similar to that used in the tensile test was produced, cut to 40 mm×10 mm, and used for measurement. Using an "EXSTAR DMS6100" produced by Hitachi High-Tech Science Corporation, a storage modulus (E') and a loss modulus (E'') were measured by, in accordance with JIS K7244-4:1999, measuring viscoelasticity in a nitrogen air flow (300 mL/min), in a temperature range of from -50° C. to 300° C. at a measurement frequency of 1 Hz, a heating rate of 3° C./min, a strain of 0.1% (20 m) in a tensile mode. The value of (E'')/(E') was calculated as the loss tangent (tan δ).

(180) <Heat Deformation Temperature (HDT)>

(181) Using the polyimide resin composition produced in each of the examples, molded articles of 80 mm×10 mm×thickness 4 mm were produced and used for measurement by the method described later.

(182) For the measurement, a flatwise test was performed in accordance with JIS K7191-1, 2: 2015. Specifically, the heat deformation temperature was measured at distance between fulcrums of 64 mm, a load of 1.80 MPa, and a heating rate of 120° C./hour using an HDT test instrument "Auto-HDT3D-2" (produced by Toyo Seiki Seisaku-Sho, Ltd.).

Production Example 1 (Production of Crystalline Thermoplastic Polyimide Resin 1)

(183) 500 g of 2-(2-methoxyethoxy)ethanol (produced by Nippon Nyukazai Co., Ltd.) and 218.12 g (1.00 mol) of pyromellitic dianhydride (produced by Mitsubishi Gas Chemical Company, Inc.) were introduced in a 2 L separable flask equipped with a Dean-Stark apparatus, a Liebig condenser tube, a thermocouple, and a four-paddle blade. After creation of a nitrogen flow, the mixture was agitated at 150 rpm so as to become a homogeneous suspended solution. On the other hand, 49.79 g (0.35 mol) of 1,3-bis(aminomethyl)cyclohexane (produced by Mitsubishi Gas Chemical Company, Inc., cis/trans ratio=7/3) and 93.77 g (0.65 mol) of 1,8-octamethylenediamine (produced by Kanto Chemical Co., Inc.) were dissolved in 250 g of 2-(2-methoxyethoxy)ethanol with a 500 mL beaker, thereby preparing a mixed diamine solution. This mixed diamine solution was added into the suspended solution gradually with a plunger pump. Heat was generated due to the drop addition, but the internal temperature was adjusted to be within the range of 40 to 80° C. The dropwise addition of the mixed diamine solution was carried out in a nitrogen flow state over the whole period. The number of rotations of the agitation blade was set to 250 rpm. After the completion of the dropwise addition, 130 g of 2-(2-methoxyethoxy)ethanol and 1.284 g (0.010 mol) of n-octylamine (produced by Kanto Chemical Co., Inc.) as an end capping agent were added thereto, and the mixture was further agitated. At this stage, a pale yellow polyamic acid solution was obtained. Next, the agitation speed was set to 200 rpm, and the polyamic acid solution in the 2 L separable flask was then heated to 190° C. In this heating process, the deposition of a polyimide resin powder and dehydration associated with imidization were confirmed at a solution temperature of from 120 to 140° C. The solution was kept at 190° C. for 30 minutes, then allowed to cool to room temperature, and filtered. The obtained polyimide resin powder was washed and filtered with 300 g of 2-(2-methoxyethoxy)ethanol and 300 g of methanol, and then dried in a dryer at 180° C. for 10 hours to obtain 317 g of a powder of crystalline thermoplastic polyimide resin 1 (hereinafter also simply referred to as "polyimide resin 1").

(184) The measurement of the IR spectrum of polyimide resin 1 showed the characteristic absorption of an imide ring $\nu(\text{C}=\text{O})$ observed at 1768 and 1697 ($\text{cm}.\text{sup.}-1$). The logarithmic viscosity was 1.30 dL/g, T_m was 323° C., T_g was 184° C., T_c was 266° C., the exothermic amount of crystallization was 21.0 mJ/mg, the crystallization half-time was 20 seconds or less, and M_w was 55,000.

(185) Table 1 shows the composition and evaluation results of the polyimide resin 1 of Production Example 1. The values expressed in mol % of the tetracarboxylic acid component and the diamine component in Table 1 are values calculated from the charged amount of each component in production of the polyimide resin.

(186) TABLE-US-00001 TABLE 1 Tetracarboxylic Diamine acid component component Exothermic (mol % in total (mol % in amount of Half tetracarboxylic total diamine crystallization crystallization acid component) component) (1)/{(1) + (2)} T_m T_g T_c ΔH_m time PMDA 1,3-BAC OMDA (mol %)*.sup.1 (° C.) (° C.) (° C.) (mJ/mg) (seconds) M_w Production Polyimide 100 35 65 35 323 184 266 21.0 <20 55,000 Example 1 resin 1 *.sup.1 Content ratio (mol %) of repeating structural unit of formula (1) with respect to total of repeating structural unit of formula (1) and repeating structural unit of formula (2) in polyimide resin 1

(187) The abbreviations in Table 1 are as follows. PMDA; pyromellitic dianhydride 1,3-BAC; 1,3-bis(aminomethyl)cyclohexane OMDA; 1,8-octamethylenediamine

Examples 1 to 9 and Comparative Example 2 (Production and Evaluation of Polyimide Resin Composition and Molded Article)

(188) The polyimide resin 1 obtained in Production Example 1 and the polyetherimide sulfone resin (“EXTEM VH1003P”, produced by Sabic Innovative Plastics) were dry-blended in the ratio shown in Table 2, and then the mixture was melt-kneaded and extruded using a co-rotating twin-screw kneading extruder (“HK-25D”, produced by Parker Corporation, screw diameter $D=25$ mm, $L/D=41$) at a barrel temperature of 370° C. and a screw rotation speed of 150 rpm. The strand extruded from the extruder was air cooled, and then pelletized with a pelletizer (“Fan Cutter FC-Mini-4/N”, produced by Hoshi Plastic Co., Ltd.). The obtained pellets were dried at 150° C. for 12 hours, and then used for injection-molding.

(189) The injection-molding was performed at a barrel temperature of 385° C., a mold temperature of 165° C., and a molding cycle of 60 seconds with an injection-molding machine (“ROBOSHOT α -S30iA”, produced by FANUC Corporation), thereby preparing a molded article having a desired shape to be used in the various evaluations.

(190) Various evaluations were performed by the methods described above using the obtained pellets and molded articles. The results are shown in Table 2.

Comparative Example 1

(191) The polyimide resin 1 obtained in Production Example 1 was kneaded and extruded at a barrel temperature of 350° C. and a screw rotation speed of 70 rpm using a Labo Plasto Mill (produced by Toyo Seiki Seisaku-Sho, Ltd.). The strand extruded from the extruder was air cooled, and then pelletized with a pelletizer (“Fan Cutter FC-Mini-4/N”, produced by Hoshi Plastic Co., Ltd.). The obtained pellets were dried at 150° C. for 12 hours, and then used for injection-molding.

(192) The injection-molding was performed at a barrel temperature of 350° C., a mold temperature of 200° C., and a molding cycle of 50 seconds with an injection-molding machine (“ROBOSHOT α -S30iA”, produced by FANUC Corporation), thereby preparing a molded article having a desired shape to be used in the various evaluations.

(193) Various evaluations were performed by the methods described above using the obtained pellets and molded article. The results are shown in Table 2.

(194) TABLE-US-00002 TABLE 2 Comparative Example Example 1 1 2 3 4 5 6 Resin (A1) Polyimide 100 90 80 70 60 50 40 compositional resin 1 makeup (parts (B1) Polyetherimide 0 10 20 30 40 50 60 by mass) sulfone resin (Extem) Heat T_m ° C. 323 321 321 321 320 321 319 properties T_g ° C. 184 185 182 185 240 239 240 T_c ° C. 266 265, 234 267, 235 268, 235 269 266 267 ΔH_m

mJ/mg 21.0 18.4 16.3 14.4 11.8 9.0 7.1 Mechanical Bending MPa 120 112 114 120 121 127 131
 properties strength Flexural GPa 2.6 2.4 2.4 2.5 2.5 2.6 2.7 modulus Tensile MPa 80 73 75 77 79
 84 86 strength Tensile GPa 2.5 2.9 2.4 3.1 2.5 3.6 2.8 modulus Tensile % 21 54 102 95 95 45 28
 fracture strain Color hue L — 72 37 37 41 43 45 42 a — 1.4 5.6 4.8 4.1 3.7 3.8 3.2 b — 21.4 10.8
 10.0 9.9 11.2 11.4 10.3 YI — 54.9 64.2 58.2 50.3 52.4 52.0 49.0 Degree of — 64.6 35.4 36.0 40.2
 42.3 43.4 41.4 whiteness Visco- Storage 10.sup.9 Pa 3.2  custom character  custom character 3.2
 custom character  custom character  custom character elasticity modulus Loss 10.sup.7 Pa 3.8
 custom character  custom character 4.6  custom character  custom character
 custom character modulus tan δ — 0.0116  custom character  custom character 0.0148
 custom character  custom character  custom character Comparative Example Example 7 8 9 2
 Resin (A1) Polyimide 30 20 10 0 compositional resin 1 makeup (parts (B1) Polyetherimide 70 80
 90 100 by mass) sulfone resin (Extem) Heat Tm ° C. 314-328, 312-323, 313, 297 — properties 296
 296 Tg ° C. 240 240 244 243 Tc ° C. 264, 248 245 245 — ΔH_m mJ/mg 4.9 2.9 1.0 — Mechanical
 Bending MPa 135 143 141 143 properties strength Flexural GPa 2.7 2.9 2.9 3.0 modulus Tensile
 MPa 88 93 91 94 strength Tensile GPa 3.2 3.7 4.0 3.5 modulus Tensile % 34 20 12 11 fracture
 strain Color hue L — 43 43 37 27 a — 3.0 3.1 3.9 4.8 b — 10.4 11.1 11.0 8.2 YI — 49.1 53.4 60.9
 68.0 Degree of — 41.6 40.5 35.9 26.0 whiteness Visco- Storage 10.sup.9 Pa  custom character
 custom character  custom character  custom character elasticity modulus Loss 10.sup.7 Pa
 custom character  custom character  custom character  custom character modulus tan δ —
 custom character  custom character  custom character  custom character * Diagonal line
 means not measured

(195) The details of each component shown in Table 2 are as follows.

(196) <Polyimide Resin (A)>

(197) (A1) Polyimide resin 1: Crystalline thermoplastic polyimide resin obtained in Production Example 1

(198) <Polyetherimide Sulfone Resin (B)>

(199) (B1) Polyetherimide sulfone resin (Extem): Polyetherimide sulfone resin containing a repeating structural unit (repeating structural unit represented by formula (4-i)) derived from 2,2-Bis[4-(2,3-dicarboxyphenoxy)phenyl]propane dianhydride and diaminodiphenyl sulfone, “EXTEM VH1003P” produced by Sabic Innovative Plastics, Tg 243° C.

(200) As shown in Table 2, the molded articles composed of the polyimide resin composition of Examples 1 to 9, which contain both the polyimide resin (A1) and the polyetherimide sulfone resin (B1), had better tensile modulus and better tensile fracture strain than the molded article of Comparative Example 1 composed only of the polyimide resin (A1) and the molded article of Comparative Example 2 composed only of the polyetherimide sulfone resin (B1). The molded articles obtained in Examples 1 to 7 had better tensile break strain values than the molded articles of Comparative Examples 1 and 2, and had higher toughness. In particular, it can be seen that the values of tensile break strain were especially improved when the mass ratio [(A1)/(B1)] of the component (A1) to the component (B1) was in the range of 50/50 to 90/10 (Examples 1 to 5) and 60/40 to 80/20 (Examples 2 to 4). On the other hand, the molded articles obtained in Examples 8 and 9 had better tensile modulus than the molded articles of Comparative Examples 1 and 2.

(201) Further, it can be seen that the polyimide resin compositions of Examples 1 to 9 had a higher degree of whiteness than that of the polyetherimide sulfone resin (B1) alone of Comparative Example 2.

(202) The molded article composed of the polyimide resin composition of Example 3 had the same storage modulus as the polyimide resin (A1) alone of Comparative Example 1, but had better loss modulus and tan δ values.

(203) In addition, using pellets of the polyimide resin compositions obtained in Examples 3, 5, and 7, the dispersion state of the polyimide resin (A1) and the polyetherimide sulfone resin (B1) in each pellet was confirmed by the following method.

(204) The pellets obtained in Examples 3, 5, and 7 were cut perpendicular to the flow direction (MD) of the pellets using a microtome (“ULTRACUT E”, produced by REICHERT-JUNG LIMITED) to produce ultra-thin pieces. After dyeing the cut surface with ruthenium tetroxide, the pieces were observed at an acceleration voltage of 30 kV and an observation magnification of 30,000 times using a field-emission scanning transmission electron microscope (FE-STEM, “Gemini SEM500”, produced by ZEISS) (FIGS. 1 to 3). In each observation image, the portions where the dye was darker were determined to be composed of the polyetherimide sulfone resin (B1), which is easily stained with ruthenium tetroxide.

(205) FIG. 1 is a micrograph of the pellets of Example 3 (mass ratio [(A1)/(B1)]=70/30), FIG. 2 is a micrograph of the pellets of Example 5 (mass ratio [(A1)/(B1)]=50/50), and FIG. 3 is a micrograph of the pellets of Example 7 (mass ratio [(A1)/(B1)]=30/70).

(206) From FIGS. 1 to 3, it can be seen that in the pellets obtained in Examples 3, 5, and 7, the polyimide resin (A1) and the polyetherimide sulfone resin (B1) are uniformly dispersed, and that a nano-level sea-island structure is formed. In the pellets of Example 3, it can be seen that the polyimide resin (A1) constitutes the “sea” of the sea-island structure, and in the pellets of Examples 5 and 7, the polyimide resin (A1) constitutes the “islands” of the sea-island structure.

Comparative Examples 3 to 5

(207) The polyimide resin 1 obtained in Production Example 1 and the polyetherimide-siloxane copolymer (b1) (“SILTEM resin STM1700”, produced by Sabic Innovative Plastics), which is a thermoplastic resin other than the component (B), were dry-blended in the ratio shown in Table 3, and then the mixture was melt-kneaded and extruded using a co-rotating twin-screw kneading extruder (“HK-25D”, produced by Parker Corporation, screw diameter D=25 mm, L/D=41) at a barrel temperature of 335° C. and a screw rotation speed of 150 rpm. The strand extruded from the extruder was air cooled, and then pelletized with a pelletizer (“Fan Cutter FC-Mini-4/N”, produced by Hoshi Plastic Co., Ltd.). The obtained pellets were dried at 105° C. for 6 hours, and then used for injection-molding.

(208) The injection-molding was performed at a barrel temperature of 370° C., a mold temperature of 165° C., and a molding cycle of 60 seconds with an injection-molding machine (“ROBOSHOT α-S30iA”, produced by FANUC Corporation), thereby preparing a molded article having a desired shape to be used in the various evaluations.

(209) The various evaluations listed in Table 3 were performed by the methods described above using the obtained molded article. The results are shown in Table 3.

(210) TABLE-US-00003

	Example 3	Example 5	Example 7	Example 3	Example 5	Example 7	Example 3	Example 5	Example 7
Resin (A1)	1	70	50	30	70	50	30	70	50
Polyimide resin	1	70	50	30	70	50	30	70	50
compositional (B1)	0	0	0	0	0	0	0	0	0
Polyetherimide	0	0	0	0	0	0	0	0	0
sulfone resin (Extem)	0	0	0	0	0	0	0	0	0
by mass (b1)	0	0	0	0	0	0	0	0	0
Polyetherimide-siloxane copolymer	0	0	0	0	0	0	0	0	0
(STM 1700)	0	0	0	0	0	0	0	0	0
HDT (load 1.80 MPa) ° C.	180	190	200	164	160	157	164	160	157
Mechanical Bending strength MPa	120	127	135	97	99	104	97	99	104
Flexural modulus GPa	2.5	2.6	2.7	1.9	2.0	2.1	1.9	2.0	2.1
Tensile strength MPa	77	84	88	62	66	70	62	66	70
Tensile modulus GPa	3.1	3.6	3.2	1.9	1.9	2.2	1.9	1.9	2.2
Tensile fracture %	95	45	34	30	45	34	30	45	34
strain	95	45	34	30	45	34	30	45	34

* Diagonal line means not measured

(211) The details of each component shown in Table 3 are as follows.

(212) <Polyimide Resin (A)>

(213) (A1) Polyimide resin 1: Crystalline thermoplastic polyimide resin obtained in Production Example 1

(214) <Polyetherimide Sulfone Resin (B)>

(215) (B1) Polyetherimide sulfone resin (Extem): Polyetherimide sulfone resin containing a repeating structural unit (repeating structural unit represented by formula (4-1)) derived from polyetherimide sulfone resin 2,2-Bis[4-(2,3-dicarboxyphenoxy)phenyl]propane dianhydride and diaminodiphenyl sulfone, “EXTEM VH1003P” produced by Sabic Innovative Plastics, Tg 243° C.

(216) <Thermoplastic Resin Other than (B)>

(217) (b1) Polyetherimide-siloxane copolymer <STM 1700>: “SILTEM resin STM1700”,

produced by Sabic Innovative Plastics

(218) As shown in Table 3, when Example 3 and Comparative Example 3 are compared, the molded article composed of a polyimide resin composition of Comparative Example 3, which used the polyetherimide-siloxane copolymer (b1) instead of the polyetherimide sulfone resin (B1), had lower values for tensile modulus and tensile fracture strain than Example 3. Further, it can be seen that the HDT and bending properties were also low.

(219) When Example 5 and Comparative Example 4 are compared, and Example 7 and Comparative Example 5 are compared, although the values for tensile fracture strain were the same, for other properties, the cases where the polyetherimide-siloxane copolymer (b1) was used showed inferior results.

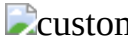
Examples 10 and 11, Comparative Examples 6 and 7 (Production and Evaluation of Polyimide Resin Compositions and Molded Article)

(220) The polyimide resin 1 obtained in Production Example 1, and of the various component shown in Table 4, the components other than carbon fiber were thoroughly mixed by dry blending. The obtained mixed powders were introduced into a co-rotating twin-screw kneading extruder ("HK-25D", produced by Parker Corporation, screw diameter D=25 mm, L/D=41), carbon fiber (Tenax-A HT P722, produced by Teijin Industries, sizing agent: polyimide-based, average fiber length: 3 mm) in the amount shown in Table 4 was side-fed, and the resulting mixture was extruded at a barrel temperature of 335° C. and a screw rotation speed of 150 rpm. A strand extruded from the extruder was cooled in air and then pelletized with a pelletizer ("Fan Cutter FC-Mini-4/N", produced by Hoshi Plastic Co., Ltd.).

(221) The obtained pellets were dried at 190° C. for 10 hours and then used in injection molding.

(222) The injection molding was performed at a barrel temperature of 350° C. and a mold temperature of 200° C., and a molding cycle of 50 seconds with an injection-molding machine ("ROBOSHOT α-S30iA", produced by FANUC Corporation), thereby preparing a molded article having a desired shape to be used in the various evaluations.

(223) The various evaluations listed in Table 4 were performed using the obtained pellets and molded articles, by the methods described above. The results are shown in Table 4.

(224) TABLE-US-00004 TABLE 4 Example Comparative Example 3 10 11 6 7 Resin (A1) Polyimide resin 1 70 70 70 70 70 compositional (B1) Polyetherimide 30 30 30 0 0 makeup (parts sulfone resin (Extm) by mass) (b1) Polyetherimide- 0 0 0 30 30 siloxane copolymer (STM 1700) Carbon fiber content (% by mass) 0 30 50 30 50 HDT (load 1.80 MPa) ° C. 180 224 220 211 186 Mechanical Bending strength MPa 120 266 261 213 145 properties Flexural modulus GPa 2.5 17.2 27.3 19.7 26.7 Tensile strength MPa 77 156  132  Tensile modulus GPa 3.1 19.8  21.2  Tensile fracture % 95 1  1  strain * Diagonal line means not measured

(225) In Table 4, the molded article of Example 10 composed of a polyimide resin composition containing carbon fiber, has better tensile strength, tensile modulus, and HDT than the molded article of Example 3 that did not contain carbon fiber. Further, it can be seen that the molded articles of Examples 10 and 11 had better bending strength and HDT than the molded articles of Comparative Examples 6 and 7, which used the polyetherimide-siloxane copolymer (b1) as a comparative substance instead of a polyetherimide sulfone resin.

INDUSTRIAL APPLICABILITY

(226) The polyimide resin composition and molded article of the present invention have excellent heat resistance and bending properties and good tensile properties, and therefore are expected to be developed in applications where importance is placed on impact resistance, vibration control, and the like. For example, the polyimide resin composition and molded article of the present invention can be applied to applications such as sliding members such as gears and bearings, cutting members, structural members such as robot arms, winding coating materials such as electric wires,

screws, nuts, packings, speaker diaphragms, reflectors, fifth generation (5G) mobile communication system related members, various films, and the like.

Claims

1. A polyimide resin composition comprising a crystalline thermoplastic polyimide resin (A) and a polyetherimide sulfone resin (B), wherein the polyimide resin (A) comprises a repeating structural unit represented by the following formula (1) and a repeating structural unit represented by the following formula (2), and a content ratio of the repeating structural unit of the formula (1) with respect to the total of the repeating structural unit of the formula (1) and the repeating structural unit of the formula (2) is from 20 to 70 mol %: ##STR00016## wherein R.sub.1 represents a divalent group having from 6 to 22 carbon atoms containing at least one alicyclic hydrocarbon structure; R.sub.2 represents a divalent chain aliphatic group having from 5 to 16 carbon atoms; and X.sub.1 and X.sub.2 each independently represent a tetravalent group having from 6 to 22 carbon atoms containing at least one aromatic ring, and wherein a mass ratio of the polyimide resin (A) to the polyetherimide sulfone resin (B), [(A)/(B)], is from 1/99 to 25/75.
2. The polyimide resin composition according to claim 1, wherein in the polyimide resin (A), a content ratio of the repeating structural unit of the formula (1) with respect to a total of the repeating structural unit of the formula (1) and the repeating structural unit of the formula (2) is from 20 mol % to less than 40 mol %.
3. The polyimide resin composition according to claim 1, wherein the polyetherimide sulfone resin (B) has a glass transition temperature of at least 230° C.
4. The polyimide resin composition according to claim 1, wherein the polyetherimide sulfone resin (B) comprises a repeating structural unit represented by the following formula (4): ##STR00017## wherein R.sub.4 represents a divalent group having from 6 to 22 carbon atoms comprising —SO.sub.2—, and X.sub.4 represents a tetravalent group having from 12 to 40 carbon atoms comprising at least one aromatic ring and an ether bond.
5. The polyimide resin composition according to claim 4, wherein in the repeating structural unit represented by the formula (4), R.sub.4 is a divalent group represented by the following formula (R4-1): ##STR00018## wherein R.sub.41 and R.sub.42 are each independently an alkyl group having from 1 to 4 carbon atoms, an alkenyl group having from 2 to 4 carbon atoms, or an alkynyl group having from 2 to 4 carbon atoms, m.sub.41 and m.sub.42 are each independently an integer of from 0 to 2, and p41 and p42 are each independently an integer of from 0 to 4.
6. The polyimide resin composition according to claim 4, wherein in the repeating structural unit represented by the formula (4), X.sub.4 is a tetravalent group represented by the following formula (X-7): ##STR00019## wherein L.sub.41 is a divalent group having from 4 to 28 carbon atoms.
7. The polyimide resin composition according to claim 6, wherein in the formula (X-7), L.sub.41 is a divalent group represented by any of the following formulas (X-7a) to (X-7e): ##STR00020## wherein R.sub.43 represents an alkyl group having from 1 to 4 carbon atoms, and p43 represents an integer of from 0 to 4.
8. The polyimide resin composition according to claim 1, further comprising a reinforcement fiber.
9. A molded article comprising the polyimide resin composition of claim 1.
10. A polyimide resin composition comprising a crystalline thermoplastic polyimide resin (A) and a polyetherimide sulfone resin (B), wherein the polyimide resin (A) comprises a repeating structural unit represented by the following formula (1) and a repeating structural unit represented by the following formula (2), and a content ratio of the repeating structural unit of the formula (1) with respect to the total of the repeating structural unit of the formula (1) and the repeating structural unit of the formula (2) is from 20 to 70 mol %: ##STR00021## wherein R.sub.1 represents a divalent group having from 6 to 22 carbon atoms comprising at least one alicyclic hydrocarbon structure; R.sub.2 represents a divalent chain aliphatic group having from 5 to 16 carbon atoms;

and X.sub.1 and X.sub.2 each independently represent a tetravalent group having from 6 to 22 carbon atoms comprising at least one aromatic ring, and wherein a mass ratio of the polyimide resin (A) to the polyetherimide sulfone resin (B), [(A)/(B)], is from 80/20 to 99/1.

11. The polyimide resin composition according to claim 10, wherein in the polyimide resin (A), a content ratio of the repeating structural unit of the formula (1) with respect to a total of the repeating structural unit of the formula (1) and the repeating structural unit of the formula (2) is from 20 mol % to less than 40 mol %.

12. The polyimide resin composition according to claim 10, wherein the polyetherimide sulfone resin (B) has a glass transition temperature of at least 230° C.

13. The polyimide resin composition according to claim 10, wherein the polyetherimide sulfone resin (B) comprises a repeating structural unit represented by the following formula (4):

##STR00022## wherein R.sub.4 represents a divalent group having from 6 to 22 carbon atoms comprising —SO.sub.2—, and X.sub.4 represents a tetravalent group having from 12 to 40 carbon atoms comprising at least one aromatic ring and an ether bond.

14. The polyimide resin composition according to claim 13, wherein in the repeating structural unit represented by the formula (4), R.sub.4 is a divalent group represented by the following formula (R4-1): ##STR00023## wherein R.sub.41 and R.sub.42 are each independently an alkyl group having from 1 to 4 carbon atoms, an alkenyl group having from 2 to 4 carbon atoms, or an alkynyl group having from 2 to 4 carbon atoms, m.sub.41 and m.sub.42 are each independently an integer of from 0 to 2, and p41 and p42 are each independently an integer of from 0 to 4.

15. The polyimide resin composition according to claim 13, wherein in the repeating structural unit represented by the formula (4), X.sub.4 is a tetravalent group represented by the following formula (X-7): ##STR00024## wherein L.sub.41 is a divalent group having from 4 to 28 carbon atoms.

16. The polyimide resin composition according to claim 15, wherein in the formula (X-7), L.sub.41 is a divalent group represented by any of the following formulas (X-7a) to (X-7e): ##STR00025## wherein R.sub.43 represents an alkyl group having from 1 to 4 carbon atoms, and p43 represents an integer of from 0 to 4.

17. The polyimide resin composition according to claim 10, further comprising a reinforcement fiber.

18. A molded article comprising the polyimide resin composition of claim 10.
